

Dongwook Kwon

AI ENGINEER · AGENTIC AI SYSTEMS

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Summary

Highly accomplished AI Research Engineer specializing in Large Language Model (LLM) optimization and Multi-Agent Systems (MAS), with a proven track record of bridging advanced academic research and high-impact industrial applications. Demonstrated research excellence through publications in top-tier natural language processing venues, including an Oral presentation at EMNLP 2025 and a peer-reviewed paper at ACL 2026 focusing on cost-aware LLM cascades. Inventor of multiple registered patents in AI-driven anomaly detection and a key technical lead who has successfully delivered mission-critical autonomous agent systems for global enterprise leaders, including Hyundai Motor Company and LG Electronics. Poised to significantly advance the United States' technological edge in critical and emerging fields by engineering scalable, reliable, and secure autonomous AI architectures.

Patents

Apr 2026 **Method for Implantable Medical Device to Detect Anomaly in Real Time**, KR 10-2961984

Republic of Korea

Aug 2025 **GPT API-Based Network Anomaly Detection**, KR 10-2848814

Republic of Korea

Publications

JOURNAL ARTICLES

- Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.
- D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," under review at *Data Mining and Knowledge Discovery (Springer Nature)*, 2025.

CONFERENCE PAPERS

- R. Chang[†], **D. Kwon**[†], J. Lee[†], and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) — Industry Track (Poster)*, 2026.
[†]Equal contribution.
- J. Lee[†], R. Chang[†], **D. Kwon**[†], H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025.
[†]Equal contribution.
- D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. **Best Poster Award — Silver.**
- D. Kwon**, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. **Outstanding Paper Award.**

Experience

Suresoft Technologies

Seongnam-si, Korea

AI ENGINEER, INTELLIGENT AGENT TECHNOLOGY TEAM

Dec. 2025 - Present

- Developing an AI validation system utilizing AI agent-based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Building agentic AI systems to enable systematic, scalable, and reliable AI verification processes.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University

Seoul, Korea

RESEARCH ASSISTANT

Aug. 2021 - Dec. 2025

- Researched deep-learning algorithms for anomaly detection in time-series data (ECG, cyber-physical systems).
- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.
- Published in EMNLP 2025 (Oral) and ACL 2026; co-author on two granted Republic of Korea patents.

LG Electronics Canada (with University of Toronto)

Toronto, ON, Canada

VISITING RESEARCHER

Jan. 2025 - Jul. 2025

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

Qualcomm Institute, University of California San Diego

AI DEVELOPMENT PROJECT PARTICIPANT, SUMMER PROGRAM

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

San Diego, CA, USA

Jul. 2022 - Aug. 2022

Education

Kwangwoon University

MASTER OF SCIENCE IN COMPUTER ENGINEERING

- GPA: 4.33/4.5 (98.1%)

Seoul, Korea

Mar. 2024 - Feb. 2026

University of Toronto

GRADUATE RESEARCH PROGRAM IN MECHANICAL AND INDUSTRIAL ENGINEERING

- GPA: 3.68/4.0

Toronto, ON, Canada

Jan. 2025 - Jul. 2025

Kwangwoon University

BACHELOR OF SCIENCE IN ELECTRONICS AND COMMUNICATIONS ENGINEERING

- GPA: 3.41/4.5 (88.1%)

Seoul, Korea

Mar. 2019 - Feb. 2024

Honors & Awards

MAJOR AWARDS

Aug. 2022 **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego

United States

Dec. 2021 **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries

South Korea

ACADEMIC RECOGNITION

Nov. 2025 **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers

South Korea

Nov. 2024 **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024)

South Korea

Nov. 2023 **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering

South Korea

Funded Research Projects

LG Electronics Canada

AGENTIC AI EVALUATION FRAMEWORK

- Developed an evaluation framework for agentic AI systems using LLMs to assess performance.

Toronto, ON, Canada

Jan. 2025 - Present

Sejong University

AI-BASED INTRUSION DETECTION AND ATTACK PACKET CRAFTING TECHNOLOGIES FOR APR1400

- Developed an AI defense system to protect nuclear power plants from cyber attacks.

Seoul, Korea

Jul. 2022 - Present

Ministry of Science and ICT

QUADRUPEL ROBOT USING VISION SLAM WITH A DEPTH CAMERA

- Developed an autonomous quadruped robot using a navigation algorithm with SLAM and a depth camera.

Seoul, Korea

Apr. 2022 - Nov. 2022

Qualcomm Institute, University of California San Diego

PERSONALITY-BASED DRUG ADDICTION PREDICTION SYSTEM

- Developed a personality-based drug addiction prediction system using a machine learning model.

San Diego, CA, USA

Jul. 2022 - Aug. 2022

Kookmin University

DEVELOPMENT OF ACTIVE KILL-SWITCH AND BIOMARKER-BASED DEFENSE SYSTEM FOR LIFE-THREATENING IOT MEDICAL DEVICES

- Developed a security system for heart implant devices using an anomaly detection algorithm.

Seoul, Korea

Sep. 2021 - Dec. 2022

Ministry of Oceans and Fisheries

DEVELOPMENT OF A SMART LOGISTICS SYSTEM USING COMPUTER VISION

- Developed a logistics detection system incorporating location tracking and stack-layer identification for containers.

Seoul, Korea

Apr. 2021 - Nov. 2021

Kwangwoon University

SMART KITCHEN IOT SYSTEM UTILIZING COMPUTER VISION AND DEEP LEARNING

- Developed a cloud IoT platform featuring a system for classifying groceries in the fridge and an OCR receipt program.

Seoul, Korea

May 2021 - Oct. 2021